LESSON 2.11

Add and Subtract Decimals

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can add and subtract decimals by lining up the place values and the decimal points.

WHY THIS MATTERS The space station's main fuel tank reads 128.75 liters. A resupply shuttle delivers 46.8 liters of fuel, but the thrusters used 19.35 liters during docking. What is the fuel level after both events?

REMEMBER

Add and Subtract Decimals

Adding & Subtracting Decimals A decimal is a number with a point that shows parts smaller than one whole. To add or subtract decimals, you line up the decimal points so the same place values match.

1. Line up the decimal points vertically by place value.
2. Insert placeholder zeros so each number has the same number of decimal digits.
3. Add or subtract right-to-left, regrouping as needed, and drop decimal straight down

Words: **Decimal *(Decimal)*** **Place value *(Valor posicional)*** **Tenths *(Décimas)*** **Hundredths *(Centésimas)*** **Annex zeros *(Agregar ceros)***

1 **Circle the letter of the best answer. Show how you know.**

A data set has one very large outlier. Which measure of centre does it pull on MORE?

(A) The median
(B) Neither one
(C) The mean
(D) Both equally

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

House prices on a street are $180,000, $190,000, $200,000, $210,000 and $2,000,000. Which measure best describes a typical price?

(A) The mean
(B) The median
(C) The range
(D) The largest value

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A data set is roughly symmetric with no outliers. Which measure of centre is a good choice?

(A) The range
(B) Only the maximum
(C) The mean
(D) Neither the mean nor the median

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Test scores are 70, 72, 75, 74 and 12. Find the median.

(A) 60.6
(B) 72
(C) 74
(D) 75

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

5 Circle the letter of the best answer. Show how you know.

A runner's times for three laps are 2.45 minutes, 2.6 minutes, and 2.38 minutes. She wants to find her total time to see if she beat her personal record of 7.5 minutes. 2.45 + 2.60 + 2.38 = ?

(A) 6.43
(B) 7.143
(C) 7.43
(D) 7.53

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

6 Circle the letter of the best answer. Show how you know.

A runner's times for three laps are 2.45 minutes, 2.6 minutes, and 2.38 minutes. She wants to find her total time to see if she beat her personal record of 7.5 minutes. Why is it safe to rewrite 2.6 as 2.60?

(A) It makes the number ten times bigger
(B) Adding a zero at the end of a decimal does not change its value
(C) It rounds the number up
(D) Because 6 and 60 use the same digit

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does a number line help you see what happens when you add or subtract decimals?

SENTENCE STARTER When I add ___ to ___, I move ___ to the right on the number line and land on ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 Write the letter of the matching item on each line.

Match each receipt or log expression to its correct sum or difference.

_____ 18.4 + 6.95	(A) 25.35
_____ 22.5 − 9.8	(B) 12.7
_____ 7.06 + 3.4	(C) 10.46
_____ 40 − 13.55	(D) 26.45
_____ 16.2 − 9.65	(E) 6.55

2 Write the name of the correct group on each line.

Estimate each receipt total by rounding to the nearest whole dollar, then sort: was the exact result more than $20 or less than $20?

Groups: More Than $20 | Less Than $20

_____ $13.40 + $9.75

_____ $30.00 − $4.85

_____ $11.60 + $8.99

_____ $12.75 − $3.40

_____ $6.50 + $7.25

_____ $25.10 − $7.45

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A runner's times for three laps are 2.45 minutes, 2.6 minutes, and 2.38 minutes. She wants to find her total time to see if she beat her personal record of 7.5 minutes. Did she beat her personal record of 7.5 minutes?

(A) No, because 7.43 > 7.5

(B) Yes, because 7.43 < 7.5

(C) No, because 7.43 and 7.5 are equal

(D) You cannot compare them

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Case File: A crew member's snack receipt shows $27.60 for dried fruit and $15.45 for protein bars. What is the total on the receipt?

(A) $42.05

(B) $42.95

(C) $43.05

(D) $43.15

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Case File: The oxygen gauge dropped from 9.6 units to a new reading after the crew used 4.75 units. How much oxygen is left?

(A) 4.85

(B) 4.95

(C) 5.15

(D) 5.85

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Case File: A supply log started the shift with $48.00 in the station snack fund. By the end, $31.27 had been spent. How much is left in the fund?

(A) $16.73
(B) $16.83
(C) $17.27
(D) $17.73

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does a number line help you see what happens when you add or subtract decimals?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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